Manuscript outline: what each paragraph should contain

enceladus_DSM `studies/keff_sintering/manuscript_outline.tex`

2026-10-12, against the draft of the same date

Framing. Dry snow metamorphism is the story; the model covers one part of it, sintering by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach. Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to $-5\,^{\circ}\mathrm{C}$, air at 1 atm, sealed pores, 50 μm grains, 30 days, two dimensions.

How to read this. This is a plan of content: each numbered item is one paragraph and lists what it should say, with the numbers. Line numbers (L first–last) are those of the draft of 2026-10-12 in `Literature/` (36 pages, 669 lines). **[written]** the paragraph exists and says this; **[to write]** it does not exist yet; **[fix]** one specific thing in it is wrong; **[trim]** it can be shorter. Boxes mark where a figure or table goes.

Sections 4 and 5 of the draft are treated as notes. They were written for an earlier model and an earlier set of results. Nothing they say about results is carried over. They are used only as a source of points about the literature, marked **[from the notes]** , each to be checked against its source.

Suggested citations follow each paragraph, with the key in your `.bib`. Most are the ones the old Discussion used. I have matched them by title only; check each claim against the paper before it goes in.

Decisions already made, which this plan follows

Collected from the work of 2026-09-08 to 2026-10-12: the session log, the campaign record and the manuscript plan.

- **Framing** (10-08, 10-10). Laboratory-observable sintering by vapor transport, inside the dry-snow-metamorphism story. The Introduction and title stay as written; scope caveats go in the Methods. No timescales for Enceladus temperatures; the timescale map is dropped. Advisor's comments still to come.
- **Comparisons** (10-08). Not natural snow (angular, more porous, loaded). The analogues are unloaded ice-sphere and ice-powder samples: Molaro et al., Demmenie et al., Choukroun et al. From the snow literature only functional forms carry over, never magnitudes.
- **Kinetics** (09-22, 10-09). α_c constant in temperature; the thin-interface term always on; aggregates stated as $\alpha_c = 1.020 \times 10^{-3}$. No further simulations for this paper.
- **Two-grain argument** (09-09, 10-02). Rests on what the vapor route does and does not reproduce, and on exponents; no Thomas or Kingery runs. Molaro's conventions: neck *width*, time anchored on the first measurement. The $-20\,^{\circ}\mathrm{C}$ series is the benchmark.

- **Conductivity** (09-23). Tensor law in the interface; the cell problem is a steady problem on a frozen microstructure and says nothing about the transient model. Two dimensions only, so no absolute k_{eff} is claimed.
- **Clock and normalization** (10-08, 10-12). The clock is τ_{sub}; curves are normalized at $\theta_r = 30$. The first day depends on the interface parameter, so a "rise from day 1" is never quoted as a property of the ice.
- **Notation** (09-28 to 09-30). ϕ phase field, φ porosity, ζ indicator, χ curvature; subscripts "ice" and "air", not i and a; no index notation; a single dot for every contraction; the vector $\mathbf{t}$ named as in Auriault et al.
- **Citations** (09-23). Only work whose setting matches ours; every DOI checked against Crossref.
- **Words.** Desublimation ("deposition" is plume fallout); "vapor"; ε is the interface parameter, not the interface width.

Where the draft stands

Section	State
Key points, Abstract, Summary	to write (placeholders)
1 Introduction	written; small fixes and trims
2.1–2.3 Model, kinetics, homogenization	written; four definitions to add
2.4, 2.5 Numerics, simulation design	to write (material exists in Section 3)
3 Grain pairs	written, both pairs; numbers and one correction to add
4 Aggregates	written; numbers to add, one paragraph to correct
5 Discussion	to write; plan below
6 Conclusions	to write

New in this draft: the temperature table (Table 2) is in and the table notes resolve; the saturated-pair paragraph is written (L339–353); the Discussion has a new paragraph on the Molaro comparison (L436–453).

Section titles

Descriptive headings in place of "Methods" and "Results"; the Results become two sections. The rest of this outline uses the new titles.

Draft	Suggested
1 Introduction	1 Introduction
2 Methods	2 Modeling Approach
2.1 Phase-Field Model For Dry Snow Metamorphism	2.1 Phase-Field Model of Dry Snow Metamorphism
2.2 Gibbs-Thomson Condition	2.2 Interface Kinetics and the Sintering Time Scale
2.3 Homogenizing for Effective Thermal Conductivity	2.3 Effective Thermal Conductivity by Homogenization
–	2.4 Numerical Implementation
–	2.5 Simulation Design: Grain Pairs and Aggregates
3 Results	–
3.1 Phase-Field Model Validation	3 Neck Growth Between Two Ice Grains
	3.1 Comparison With Unsaturated Experiments
	3.2 Growth at Saturation
3.2 Aggregate Sintering and Effective Thermal Conductivity	4 Thermal Conductivity of Sintering Aggregates
	4.1 Microstructural Evolution
	4.2 Temperature Rescales Time
	4.3 Conductivity as a Function of Specific Surface Area
4 Discussion	5 Discussion
4.1 What vapor transport alone reproduces	5.1 What Vapor Transport Alone Reproduces
4.2 SSA is a universal predictor of $\mathbf{K}_{\text{eff}}$	5.2 Temperature as a Rescaling of Time
	5.3 Specific Surface Area as the State Variable
	5.4 What a Laboratory Could Measure
	5.5 Implications for Plume Deposits on Enceladus
	5.6 Limitations
5 Conclusion	6 Conclusions

"SSA is a universal predictor" claims more than the result: the power law holds for $\varphi \leq 0.375$ and porosity sets the level.

Build check (LaTeX warnings)

From the Overleaf project uploaded on 2026-10-11 (the newer draft came as a PDF only). As uploaded: **561 overfull-box warnings**, 144 underfull, 19 undefined citation keys, 8 undefined references. With the changes below, tested on a copy: **0 overfull, 0 underfull, 0 undefined citations**; the undefined references remain because their targets do not exist.

1. **554 of the overfull warnings: every line of body text.** The class sets ragged-right text with a 24 pt margin and only 2 em of slack (`agujournal2019.cls`, lines 113–115). TeX cannot break paragraphs within that slack, gives up, and lets each line run into the margin. Immediately after `\begin{document}`:

   ```
   \setlength{\RaggedRightRightskip}{0pt plus 4em}\RaggedRight
   \hbadness=10000 \vbadness=10000
   ```

   The first line removes the overfull warnings and leaves the page count where it is (37 pages against 38). The second hides the underfull warnings, which come from figure captions and page breaks and are harmless. Keeping the class's 24 pt margin (`24pt plus 4em`) also works and costs two pages.

2. **The other 7: the figures.** Each is included at 170 mm; the text block is 5.5 in (139.7 mm), so each sticks out 86 pt to the right. To keep the true size and center it on the text block:

   ```
   \makebox[\textwidth][c]{\includegraphics[width=170mm]{...}}
   ```

   or use `width=\textwidth` in the draft and supply the 170 mm files at submission.

3. **Table 2 was taller than the page** by 168 pt ("Float too large"). Cause: in draft mode the class gives `\footnotesize` a 20 pt line spacing, so the tables stayed double-spaced. Fixed in `tables.tex` (2026-10-11): the tables set their size with `\fontsize{8}{10}`. Table 2 now takes half a page. Re-upload `tables.tex`; nothing changes in the manuscript source.

4. **19 undefined citation keys**, all in Sections 4–5. The text uses an older key style; every one has an entry in the `.bib` under a new key. Find and replace:

`bouvet_snow_2022`	`bouvetSnowEquiTemperatureMetamorphism2022`
`braun_rigorous_2024`	`braunRigorousApproachSpecific2024`
`byrne_radar_2024`	`byrneRadarAttenuationSubsurface2024`
`calonne_numerical_2011`	`calonneNumericalExperimentalInvestigations2011`
`calonne_study_2014`	`calonneStudyTemperatureGradient2014`
`calonne_thermal_2019`	`calonneThermalConductivitySnow2019`
`fourteau_impact_2021`	`fourteauImpactWaterVapor2021`
`gallet_measurement_2009`	`galletMeasurementSpecificSurface2009`
`granger_orientation_2021`	`grangerOrientationSelectiveGrain2021`
`jaumann_distribution_2008`	`jaumannDistributionIcyParticles2008`
`kaempfer_phase-field_2009`	`kaempferPhasefieldModelingDry2009`
`libbrecht_measurements_2013`	`libbrechtMeasurementsSurfaceAttachment2013`
`libbrecht_physical_2017`	`libbrechtPhysicalDynamicsIce2017`
`matzl_measuring_2006`	`matzlMeasuringSpecificSurface2006`
`molaro_microstructural_2019`	`molaroMicrostructuralEvolutionWater2019`
`proksch_density_2015`	`prokschDensitySpecificSurface2015`
`schneebeli_tomography_2004`	`schneebeliTomographyTemperatureGradient2004`
`spencer_enceladus_2009`	`spencerEnceladusActiveCryovolcanic2009`
`spencer_enceladus_2013`	`spencerEnceladusActiveIce2013`

   Check `fourteau_impact_2021`: the `.bib` also has `fourteauMacroscopicWaterVapor2021`.

5. **Undefined references.** The table notes are resolved in the new draft. Left: six in the old Discussion, to figures and sections that no longer exist; they go when that text is replaced.

6. **Unused files in the project.** `Figure8_timescales.pdf` (dropped); `Figure7_state_law_allphi.pdf`; `agujournaltemplate.tex`; `si_template_2019.tex` (needed later for the supplement).

Front matter

- **Title.** As is, without "Version 2".
- [to write] **Key points** (three, each at most 140 characters).
  1. For a given ice-grain packing, temperature from −40 to −5 °C only rescales the time over which sintering raises conductivity.
  2. Effective conductivity follows specific surface area by one power law in well-connected packings; porosity sets its level.
  3. Vapor transport alone reproduces part of the neck growth measured between ice grains in the laboratory.

- [to write] **Abstract** (at most 250 words). Sentence by sentence:
  1. Setting: porous ice on icy bodies and on Earth; its conductivity depends on microstructure, which sintering changes.
  2. Gap: how conductivity evolves as grains sinter without compaction, and what temperature and porosity do to it.
  3. Method: a phase-field model of sublimation and desublimation with vapor and heat transport; conductivity by periodic homogenization.
  4. Grain pairs: against Molaro et al., the vapor route gives 41% of the measured neck growth at $-20\,^{\circ}\mathrm{C}$; at saturation the neck grows as $t^{0.21}$.
  5. Aggregates: 125 simulations, five porosities, five temperatures, 30 days.
  6. Result 1: for a given packing, conductivity against specific surface area is the same at every temperature within 0.3%; temperature sets the rate.
  7. Result 2: for porosity up to 0.375, conductivity scales as $\mathrm{SSA}^{-0.80}$; porosity sets the level.
  8. Meaning: SSA, which can be measured, tracks the conductivity of a sintering deposit; what a laboratory test would look like.
- [to write] **Plain Language Summary** (at most 200 words). The first half of the draft's summary (L33–43) can stay. Then: what was simulated; temperature changes how fast, not what happens; surface area tells you the conductivity; why it matters for plume deposits.

1 Introduction

Kept as drafted; no reordering. Each paragraph and what it carries:

1. **What dry snow metamorphism is** (L57–68). [written] Definition; everyday examples; that it changes bulk properties. [trim] The examples return in paragraph 2; keep them in one.
2. **Why it matters on Earth** (L69–81). [written] Avalanches, firn, conductivity of snowpacks.
3. **Icy bodies and Enceladus** (L82–101). [written] Plume, grain size, where sintering is expected, strength changes. [trim] Last sentence (L99–101) is circular. [fix] "inhabit" → harbor.
4. **Sintering as part of metamorphism** (L102–116). [written] Neck growth, the two routes, earlier work, Molaro, Choukroun. [fix] "deposits" → desublimates. Add one clause: Demmenie et al. is the second comparison used here.

   Figure 1 [written] placed. [fix] Caption and figure say "neck of radius $R(t)$"; the paper uses the neck width w.
5. **Attachment kinetics** (L117–134). [written] Supersaturation; the Wilson–Frenkel law; α_c. [trim] [fix] Shorten to those three, then end on: α_c is uncertain over two decades (10^{-3}–10^{-1}) and is held constant within each study here. As drafted the paragraph builds toward a temperature-dependent α_c through $\sigma_0(T)$, which the model does not have.
6. **Metamorphism beyond a pair; earlier models** (L135–154). [written] Kaempfer & Plapp; Bouvet; Braun; that they tune α_c. [trim] The first three sentences repeat paragraph 4.
7. **What is not known** (L155–166). [written] Bulk properties without compaction; the roles of temperature and porosity. [trim] [fix] Drop the claim that no model handles varying temperature and saturation (this paper does not close that gap) and the second telling of Bouvet and Braun.

8. **What follows** (L167–180). [written] Order of the paper. [fix] 30-day, not 28; "independent of"; delete the fragment at the end; add the saturated pair; say "compare with", not "validate"; the predictor is independent of temperature, and of porosity only in its exponent.

2 Modeling Approach

Draft: 2 Methods.

Opening (L182–189). [written] [trim] Lineage (Kaempfer & Plapp; Moure & Fu) and that this is an application of the model. Three sentences.

2.1 Phase-Field Model of Dry Snow Metamorphism

1. **Unknowns** (L191–205). [written] Two paragraphs; the first (what a phase field is) can be one sentence. Fields ϕ_i, ϕ_a (air), T, ρ_v.
2. **Scope.** (new) [to write] Three or four sentences. Of dry snow metamorphism the model carries sintering by vapor transport: sublimation, diffusion through the pore, desublimation. Not in it: surface diffusion, grain-boundary processes, densification, overburden. Define desublimation. The laboratory analogues are therefore unloaded ice-sphere samples, not natural snow.
3. **Phase equation** (L206–212). [written] The two terms of Eq. (1a). [to write] Add: M_{sub} and α_{sub} in terms of τ_{sub} and ε (or a pointer to Moure & Fu); ε named the interface parameter, with the visible band $9.2\,\varepsilon$.
4. **Saturation vapor density** (L213–217). [written] [fix] Check the last term of Eq. (3) against the source ($k_5 \ln(T + T_0)$?). One attribution: the text cites Flatau and Wexler, Table 1 says ASHRAE.
5. **Energy equation** (L218–224). [written] Mixture rules; constant properties.
6. **Vapor equation** (L225–231). [written] Diffusivity. [to write] Add one sentence on the temporal scaling ξ_T, ξ_v that Tables 3 and 4 list.

Table 1 [written] Material properties. Cited at L230.

2.2 Interface Kinetics and the Sintering Time Scale

Draft: 2.2 Gibbs-Thomson Condition.

1. **The interface condition** (L232–242). [written] Eqs. (6), (7). [to write] Add $d_0 = \gamma m/(\rho_{\mathrm{ice}} k_B T)$ in the text.
2. **The thin-interface limit.** (new) [to write] The model recovers Eq. (6) with a kinetic coefficient $\beta_{\mathrm{sub}} + \Delta\beta$, where $\Delta\beta = a_1 a_2\, \varepsilon\, (D_T^{-1} + D_v^{-1})\, \rho_{\mathrm{ice}}/\rho_{vs}^I$ (Kaempfer & Plapp, 2009; Moure & Fu, 2024). It is 2% of β_{sub} in the aggregates and 23% in the Molaro pairs.

   Cite: Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Moure & Fu 2024 `mourePhaseFieldModelWet2024`.
3. **The time scale.** (new) [to write] $\tau_{\mathrm{sub}} = \varepsilon^2(\beta_{\mathrm{sub}} + \Delta\beta)/d_0$; the sintering age $\theta = t/\tau_{\mathrm{sub}}$. ε is the same in every aggregate run, so θ compares temperatures on one clock. The 30-day runs end at $\theta = 43$ ($-40\,^\circ$C) to 1256 ($-5\,^\circ$C). The clock is τ_{sub} (decided 2026-10-12). One sentence

to keep it honest: τ_{sub} contains ε, so the values of θ quoted are for $\varepsilon = 1$ μm; ratios of τ_{sub} between temperatures do not depend on it.

4. **The values used.** (new) [to write] α_c is constant in temperature within each study: 1.020×10^{-3} (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair). Why constant: a temperature-dependent α_c would make a temperature sweep uninterpretable.

   Cite: range of α_c: Braun et al. 2024 `braunRigorousApproachSpecific2024`; Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Libbrecht 2017 `libbrechtPhysicalDynamicsIce2017`.

Table 2 [written] Temperature-dependent quantities. [fix] Not cited in the text; cite it in paragraph 3.

2.3 Effective Thermal Conductivity by Homogenization

1. **Two scales and the microscale problem** (L244–254). [written] [fix] Periodicity is an assumption ("without loss of generality" is not true of it).
2. **Macroscale law and cell problem** (L255–263). [written] [fix] Finish the broken sentence ("referred to as co…"); Eq. (11d) needs $\nabla \mathbf{t}$.
3. **Single-domain form** (L264–276). [written] [trim] Eqs. (12)–(14) can be one sentence leading to Eq. (15).
4. **Diffuse form and the tensor law** (L277–291). [written] Eq. (16), (17); why the tensor law (first-order error removed). [to write] Add two sentences. Credit: Nicoli, Plapp & Henry (2011) define the law and show the cancellation for a planar interface at contrasts of 2 and 10; here it is applied to the cell problem at a contrast of 115 and verified on laminate and disk-array cases (supplement). Why it matters here: with a scalar law the bias reaches tens of percent and grows with ε SSA, so it would drift as the aggregate sinters. State that the cell problem is solved on frozen snapshots of ϕ.

   Cite: Nicoli, Plapp & Henry 2011 `nicoli_tensorial_2011`; Auriault 1991 `auriaultHeterogeneousMediumEquivalent1991`; Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.
5. **What is reported.** (new) [to write] Three definitions: $\mathbf{K}_{\text{eff}}$ in the diffuse form, the one evaluated; $k_{\text{eff}} = (k_{xx} + k_{yy})/2$; and SSA, how it is measured from ϕ and its units. SSA is used throughout Section 4 and is defined nowhere.

   Cite: for computing SSA from a field, possibly Flin et al. 2014 `flinComputationsSpecificSurface2014` (check that the method matches).

Figure 2 [written] placed. [fix] Not cited in the text. Caption: "red" lines are orange; add that only the x problem is shown.

2.4 Numerical Implementation

1. **Discretization and time stepping.** (new) [to write] Quadratic C^1 splines; element size $\varepsilon/\sqrt{2}$; adaptive step capped at $2\,\tau_{\text{sub}}$; conductivity sampled every step for $\theta < 11$, then at every 0.1% drop in SSA.
2. **Checks.** (new) [to write] One sentence each, pointing to the supplement: domain size (Fig. S1); interface parameter (Fig. S2), including that the first day depends on ε and later rates do not; time step and sampling (Fig. S3).

Table 3 [written] Complete parameters. [fix] Not cited in the text. *Open: keep it here, or the short version here and this one in the supplement.*

2.5 Simulation Design: Grain Pairs and Aggregates

1. **Grain pairs.** (new) [to write] (material now at L315–327 and L339–347). Axisymmetric about the line of centers; the two Molaro geometries and the equal-grain pair ($R = 84.4$ μm, initial neck width 28 μm, $-20\,^{\circ}$C, 100 h); fixed temperature and humidity at the outer boundary; how the neck width w is measured.

   Table 4 [written] Two-grain parameters. [to write] Add a column for the saturated pair. [fix] Simulated time at $-5\,^{\circ}$C is 68 min.

2. **Aggregates.** (new) [to write] (material now at L357–366). 2 mm square, periodic in both directions; circular grains, log-normal radii with mean 50 μm, placed by drop-and-roll deposition; five porosities, five independent packings each, the same packings at all five temperatures: 125 runs of 30 days; uniform temperature, vapor initially at saturation. Domain size: $L/\bar{R} = 40$, at the 1–2 mm representative volume of Calonne et al. (2011). Packings are accepted only if contacts across the periodic seam are as dense as in the interior; seeds are independent across porosity (details in the supplement).

   Cite: representative volume: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`.

3. **Why periodic, why two dimensions.** (new) [to write] Periodic: the domain stands for the interior of a deposit, no wall physics has to be invented, and the system is closed, so curvature alone drives transport. Two dimensions: cost; three-dimensional effects are discussed in Section 5.6.

3 Neck Growth Between Two Ice Grains

Draft: 3 Results (opening) and 3.1 Phase-Field Model Validation.

Section opening (L293–308). [trim] Two sentences: what the two comparisons are for. The rest repeats the Introduction.

3.1 Comparison With Unsaturated Experiments

1. **Why these experiments** (L309–314). [written]
2. **What is fitted and what is not** (L315–327). [written] in part. Include: the chamber humidity is not reported, so it is inferred at each temperature from the measured shrinkage of the large grain (0.99715 at $-20\,^{\circ}$C; 0.99674 at $-5\,^{\circ}$C); $\alpha_c = 0.1$, the top of the literature range, chosen as the fastest plausible vapor route. [fix] "The only free parameter is β_{sub}": the humidity is fitted too.
3. **Result** (L328–338). [written] panels (a), (b) in words. [to write] Add: at $-20\,^{\circ}$C the model gives 41% of the measured neck growth over 78 min; at $-5\,^{\circ}$C it gives 170% over 48 min (52.1 against 44.0 μm); the authors' caveat on the $-5\,^{\circ}$C series; panels (c), (d), with the statement that the shrinkage is what the humidity was fitted to and so is not a test. Use Molaro's conventions (neck width, time from the first measurement) and say so.

Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

Figure 3 [written] placed. [fix] Caption stops at (b); say what the colors in (a) show.

3.2 Growth at Saturation

1. **Set-up and result** (L339–353). [written] Grain size, temperature, duration, how the saturation is set, the fit window, exponent 0.21. [to write] Add: why this run (at saturation the grains do not shrink, radius +0.06%, so neck growth is isolated); the neck goes from 28 to 63 μm; Demmenie et al. measure 0.26–0.33 at saturation and about 1/7 when undersaturated; the fitted exponent rises from 0.21 to 0.23 as the window starts later. For contrast, the same fit to the two unsaturated runs of Section 3.1 gives 0.135 (−20 °C) and 0.147 (−5 °C), close to 1/7, so the model does reproduce the drop in exponent away from saturation (refitted 2026-10-12 on the final runs, after the relaxation period; `studies/molaro_2019/exponents/`).

   Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`.

   [fix] L343–344 says $\alpha_c = 10^{-3}$ puts growth in "a vapor-limited regime … governed by vapor transport". It is the reverse: at this α_c growth is limited by attachment (gas diffusion adds 0.4–2% to the resistance). That is also the regime the $t^{1/3}$ law describes, so the comparison stands.

2. *Open: whether to interpret the low exponent here or in Section 5.1. Two readings: (i) yours, mechanisms absent from the model; (ii) mine, the run is faster than the Gibbs–Thomson prediction while the neck fillet is about two interface bands across, which flattens the fit; the prediction itself gives 0.29 and the final width to 4% (`studies/molaro_2019/demmenie/prediction/`). An estimate made on 2026-10-12 supports (ii) without proving it: the run's excess over the prediction is a clean function of band over fillet radius, and removing it gives 0.27 at $\varepsilon/2$ and 0.29 at $\varepsilon/4$. Only a run at $\varepsilon/2$ would settle it.*

Figure 4 [written] placed. [fix] Caption: add temperature, grain size and the fit window (1–100 h).

4 Thermal Conductivity of Sintering Aggregates

Draft: 3.2 Aggregate Sintering and Effective Thermal Conductivity.

4.1 Microstructural Evolution

1. **The gallery** (L355–372). [written] in the second half (L366–372): point contacts become necks, small grains and sharp features disappear, the pore space rounds. [to write] Add what the colors show: vapor is undersaturated over convex ice, which sublimates, and supersaturated in necks and concavities, where it desublimates. Move the set-up (first half) to Section 2.5.

Figure 5 [written] placed, with caption.

2. **The conductivity tensor** (L373–382). [written] [trim] One sentence on how it is obtained (Section 2.3). Include: k_{xy} is zero in the mean over packings and up to 12% of k_{xx} on a single one; k_{xx}/k_{yy} is near 1 for $\varphi \leq 0.375$ and 0.79 and 0.64 at $\varphi = 0.425$ and 0.475; the scalar k_{eff} is used from here on.

4.2 Temperature Rescales Time

3. **Figure 6, one packing at five temperatures** (L383–391). **[written]** panels (b)–(d) in words. **[to write]** Add, in this order:

   - Panel (a): the packing ($\varphi = 0.325$) at four instants.
   - In time, warmer runs change faster. To give a size, use numbers that do not depend on the interface parameter: between $\theta_r = 30$ and $\theta = 331$, k_{eff} rises by a factor 1.171 ± 0.015 and SSA falls to 0.819 ± 0.007 of its value, for every packing with $\varphi \leq 0.375$. The rise from day 1 (7–12% at $-40\,^{\circ}\mathrm{C}$, 31–44% at $-5\,^{\circ}\mathrm{C}$) may be quoted only as specific to $\varepsilon = 1$ μm: doubling ε changes it by 9 points and leaves the later rate unchanged.
   - Against θ the five curves fall on one: rates follow τ_{sub} within 4%.
   - Against SSA they agree within 0.27% for every one of the 25 packings (median 0.07%). This is the result; state it as a sentence.
   - SSA falls linearly in $\log\theta$, about 0.13 per decade, after the relaxation period: the form reported for isothermal snow (Legagneux et al.; to verify before citing).
     Cite: Legagneux & Dominé 2005 `legagneuxMeanFieldModel2005`; Taillandier et al. 2007 `taillandierRateDecreaseSpecific2007`. Both are in your `.bib`; for the functional form only.
   - Why curves are normalized at $\theta_r = 30$ and not at $t = 0$: the first day depends on the interface parameter (Fig. S2).

Figure 6 **[written]** placed. **[fix]** Caption: $\lvert\sigma\rvert \leq 0.34$; define θ_r.

4.3 Conductivity as a Function of Specific Surface Area

4. **Figure 7, porosity** (L392–410). **[written]** the order (a), (b), (c), (d) and the 0.3% statement. Include:

   - (a) At $-20\,^{\circ}\mathrm{C}$, mean of five packings and their range, for each porosity. At 30 d the conductivity differs by a factor of about 3.3 between $\varphi = 0.275$ and 0.475.
   - (b) Fifteen packings ($\varphi \leq 0.375$, five each) at four temperatures, 60 curves. $-40\,^{\circ}\mathrm{C}$ is absent because those runs end at $\theta = 43$, too soon after the reference age. All follow $k_{\mathrm{eff}}/k_{\mathrm{eff},r} = (\mathrm{SSA}/\mathrm{SSA}_r)^p$ with $p = -0.80$; by porosity -0.79, -0.82, -0.78.
   - (c) The level: $k_{\mathrm{eff},r}/k_{\mathrm{ice}} = 1.37\exp(-5.01\,\varphi)$ for $\varphi \leq 0.375$, with 6.6% scatter between packings. Exponential in porosity, not linear.
   - (d) Exponent of each packing. For $\varphi = 0.425$ and 0.475 it is weaker (-0.73, -0.63) and more scattered, and those packings are anisotropic; they are shown and not fitted.
   - What it amounts to: SSA gives the relative change in conductivity; porosity gives the level; neither alone gives both.
   - One closed form, if wanted: $k_{\mathrm{eff}} = 1.37\,k_{\mathrm{ice}}\,e^{-5.0\varphi}\,(\theta/30)^{0.066}$ for $\varphi \leq 0.375$ (fit of 2026-10-06).

   [fix] Three statements to correct: "3 packings for each porosity" (five); the unfinished sentence on connectivity; and the statement that the ice "remains disconnected in the horizontal direction". The last is not true of these packings: a percolation test on all 25 (2026-10-12, `studies/keff_sintering/percolation/`) finds the ice connected across the periodic cell in both directions in every packing from 10 d on, with 92–99% of the ice in the spanning cluster.

Say instead: at $\varphi = 0.425$ and 0.475 the ice network still spans but conducts unequally in the two directions ($k_{xx}/k_{yy} = 0.79$ and 0.64) and the exponent is weaker and scattered, so these packings are shown and not fitted.

Figure 7 [written] placed. [fix] Caption for (a): mean of five packings and their range, not "a single seed". Add the arrow and the gray band. A full caption was drafted on 2026-10-11.

5. **Optional paragraph: what the scatter is.** (new) [to write] The scatter in the exponent does not fall with domain size while the scatter in the level does (11% to 4.5%), so the first belongs to the packing and the second to sampling. *Open: include here, in Section 5.3, or leave out.*

5 Discussion

Draft: 4 Discussion, treated as notes. Thirteen paragraphs. For each: the argument, the numbers that carry it, and what the notes offer about the literature.

Section opening (L412–421). [trim] Not needed; the subsections carry it.

5.1 What Vapor Transport Alone Reproduces

1. **The unsaturated pair** (L436–453). [written] This is the one paragraph already written for the current results. It says: humidity inferred from shrinkage, $\alpha_c = 0.1$, under at $-20\,^{\circ}\mathrm{C}$ and over at $-5\,^{\circ}\mathrm{C}$, the authors' caveat, surface diffusion absent. Add:

   - The numbers, 41% and 170%.
   - That 41% is not a matter of tuning. With α_c at the top of its range the neck is insensitive to the chamber humidity, and with the vapor diffusivity raised 30 or 100 times the model still reaches only 75% of the measured growth once the grain shrinkage is matched; the two cases land 0.01 μm apart (tests of 2026-09-09). The vapor route has a ceiling below the measurement.
   - What fills the gap. Swinkels & Ashby's sintering model, which Molaro et al. use, has surface diffusion contributing on the order of the vapor route at these conditions; this model has none.
   - A caution on the comparison itself: Molaro et al. present their own model's agreement as order-of-magnitude, and its vapor term assumes $\alpha_c = 1$ with no gas diffusion.
   - Why $-5\,^{\circ}\mathrm{C}$ is set aside: the authors' doubts about that series (illumination and gradient-driven sublimation, stronger near melting).

   Cite: Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`; Swinkels & Ashby 1981 `swinkelsSecondReportSintering1981`; possibly Gundlach et al. 2018 `gundlachSinteringSublimationMicrometresized2018` on sintering of micrometre ice grains (in your `.bib`; check relevance).

2. **The saturated pair, and exponents.** (new) [to write] The comparison of exponents decided on 2026-10-02, as one small table or one sentence: this model at saturation 0.21–0.23; this model at Molaro's humidity 0.135 and 0.147; Demmenie et al. 0.26–0.33 at saturation and about 1/7 undersaturated; Molaro's $-20\,^{\circ}\mathrm{C}$ data 0.23 ± 0.03 (their $-5\,^{\circ}\mathrm{C}$ series has three points and no exponent); theory 1/3. What it supports: the exponent is set by the saturation state, as

Demmenie et al. argue, and the model follows that trend while sitting below the measurements. State that the run is attachment-limited and that the grain size held. *Open: the reading of the shortfall, as in Section 3.2. Also to settle: an earlier saturated run from a tangent start gave 0.28; say which run the paper quotes and why.*

Cite: Demmenie et al. 2025 `demmenieIceSinteringSublimation2025`; Molaro et al. 2019 `molaroMicrostructuralEvolutionWater2019`.

3. α_c **as an effective parameter.** (new) **[to write]** The rates in this paper scale with $1/\alpha_c$; the relation between conductivity and SSA does not depend on it. **[from the notes]** Bouvet et al. fit $\alpha_c = (9.8 \pm 0.7) \times 10^{-4}$ to tomography at $-2\,^{\circ}\mathrm{C}$ (L482–490); Braun et al. find 10^{-3}–10^{-1}, changing as surfaces facet, and call it an effective, surface-averaged value (L490–503); Kaempfer & Plapp's validation by bubble migration (L422–435).

   Cite: Bouvet et al. 2022 `bouvetSnowEquiTemperatureMetamorphism2022`; Braun et al. 2024 `braunRigorousApproachSpecific2024`; Kaempfer & Plapp 2009 `kaempferPhasefieldModelingDry2009`; Granger et al. 2021 `grangerOrientationSelectiveGrain2021` for faceting under a gradient. The notes also cite Libbrecht 2013, 2017 `libbrechtMeasurementsSurfaceAttachment2013`, `libbrechtPhysicalDynamicsIce2017` for $\sigma_0(T)$: usable only as background on why α_c is uncertain.

5.2 Temperature as a Rescaling of Time

4. **Why the curves superpose.** (new) **[to write]** One transport route, a constant α_c, transfer limited by attachment: temperature enters only through τ_{sub}, mostly through $\rho^I_{vs}(T)$. Say plainly that in this model the superposition is close to a consequence of the construction; the numerical result is how well it holds (0.27%, 4%) with heat and vapor transport both active.

   Cite: none needed; if one, Kaempfer & Plapp 2009 for the attachment-limited limit.

5. **What would break it, and the activation energy.** (new) **[to write]** The clock's temperature dependence is an activation energy of about 50 $\mathrm{kJ\,mol^{-1}}$, that of the saturation vapor density. **[from the notes]** Choukroun et al. measure 24.3 ± 3.3 $\mathrm{kJ\,mol^{-1}}$ for strength gain and attribute it to surface self-diffusion (L613–620), at 193–243 K; the two temperature ranges overlap at 233–243 K. The difference is consistent with a second route in their samples, and an activation energy is something a laboratory can measure to tell the routes apart. A temperature-dependent α_c, a second route, or a temperature gradient would each break the superposition, which makes it a test.

   Cite: Choukroun et al. 2020 `choukroun_strength_2020`.

5.3 Specific Surface Area as the State Variable

6. **Density is not enough.** (new) **[to write]** In these runs porosity is fixed and conductivity still rises, by 17% between $\theta = 30$ and 331. **[from the notes]** Calonne et al. (2011): tensor from 30 tomographic images, a regression on density, scatter attributed to anisotropy (L507–517). Calonne et al. (2014): under a temperature gradient conductivity changes at constant density, with SSA and anisotropy (L517–529). Half the length of the notes.

   Cite: Calonne et al. 2011 `calonneNumericalExperimentalInvestigations2011`; Calonne et al. 2014 `calonneStudyTemperatureGradient2014`; Schneebeli & Sokratov 2004 `schneebeliTomographyTemperatureGradient2004` (the notes cite it for the same behavior under a gradient).

7. **The power law and where it stops.** (new) **[to write]** $p = -0.80$ for $\varphi \leq 0.375$; the level from porosity; above 0.375 the exponent weakens and the packings are anisotropic. No claim that packings of different porosity converge.

Cite: Calonne et al. 2011, 2019 `calonneNumericalExperimentalInvestigations2011`, `calonneThermalConductivitySnow2019` for conductivity against density, as the contrast.

8. **Why packings differ.** (new) [to write] Packings of one porosity differ in how well the ice is connected, and the exponent follows that (correlation -0.77 with conduction efficiency), not coordination number or anisotropy. In these packings the ice always spans the cell and the pore never does (percolation test, 2026-10-12): the pore space is a set of closed pockets, which is what "sealed pores" means here, and in two dimensions only one phase can span. What differs between packings is the quality of the ice network, not whether it exists. Absolute values are therefore not compared with three-dimensional measurements, where both phases are connected. *Open: literature thresholds for discs are not yet checked against a source; whether to show the efficiency; the Euler-characteristic series (tested on three runs) is available if a direct measure is wanted.*

5.4 What a Laboratory Could Measure

9. **Three tests.** (new) [to write] On unloaded ice-sphere samples held isothermally: (i) curves at different temperatures superpose under a time shift, and the shift gives an activation energy; (ii) conductivity against SSA is one curve; (iii) SSA falls linearly in log time. [from the notes] SSA is measured by near-infrared reflectance, gas adsorption and tomography (L562–568; Gallet, Matzl, Proksch).

   Cite: Gallet et al. 2009 `galletMeasurementSpecificSurface2009`; Matzl & Schneebeli 2006 `matzlMeasuringSpecificSurface2006`; Proksch et al. 2015 `prokschDensitySpecificSurface2015`; for isothermal SSA decay in the laboratory, Legagneux & Dominé 2005, Taillandier et al. 2007, Kaempfer & Schneebeli 2007 `kaempferObservationIsothermalMetamorphism2007`, Löwe et al. 2011 `loweInterfacialStructuralRelaxations2011` (forms only).

5.5 Implications for Plume Deposits on Enceladus

10. **What carries over.** (new) [to write] Qualitative: as a deposit sinters, SSA falls and conductivity rises together, without densification. [from the notes] Tiger Stripe temperatures up to about 180 K (Spencer); grain size increasing toward the fractures in VIMS data (Jaumann; L580–602); Molaro et al. on the dependence of sintering on temperature and grain size.

    Cite: Spencer et al. 2009 `spencerEnceladusActiveCryovolcanic2009`; Spencer & Nimmo 2013 `spencerEnceladusActiveIce2013`; Jaumann et al. 2008 `jaumannDistributionIcyParticles2008`; Molaro et al. 2019; Choukroun et al. 2020.

11. **Why it matters, and the gap.** (new) [to write] [from the notes] A low-conductivity surface layer changes subsurface temperatures and radar attenuation (Byrne; L603–612); Choukroun et al. on strength gain and fallout rates of 0.01–1 $\mathrm{mm\,yr^{-1}}$ (L613–629). Then the gap, stated plainly: the runs stop at $-40\,^{\circ}\mathrm{C}$ and use 50 μm grains in air; the moon is far colder, the grains are micron-sized, and there is no air. No timescales are given.

    Cite: Byrne et al. 2024 `byrneRadarAttenuationSubsurface2024` (a preprint DOI in the `.bib`; check whether it is published); Choukroun et al. 2020; possibly Ferrari et al. 2021 `ferrariThermalEmissionSaturns2021` on thermal emission of Saturn's icy moons (in your `.bib`; check relevance).

5.6 Limitations

12. **Model.** (new) [to write] Vapor route only; constant α_c; air-filled, sealed pores at 1 atm; conduction only and constant k_{ice}, k_{air}. [from the notes] Fourteau et al. on the contribution of

vapor transport to apparent conductivity under a gradient (L569–575; check the figure quoted); Calonne et al. (2019) on temperature-dependent conductivities (L575–579).

Cite: Fourteau et al. 2021 — your `.bib` has two, `fourteauImpactWaterVapor2021` and `fourteauMacroscopicWaterVapor2021`; the point about apparent conductivity is the first. Calonne et al. 2019 `calonneThermalConductivitySnow2019`.

13. **Numerics and geometry.** (new) [to write] Two dimensions, so relative changes are claimed and absolute conductivities are not; one grain size distribution; the first day depends on the interface parameter; pore throats narrower than the diffuse band (9.2 μm) are not resolved, which raises k_{eff} most at low porosity; the power law is established for $\varphi \leq 0.375$; rates scale with $1/\alpha_c$, whose value is uncertain over two decades.

    Cite: none needed beyond those above.

In the notes and not to be carried over: $\beta_{\text{sub}}(T)$ from Libbrecht's $\sigma_0(T)$, a single tuning and a "genuine prediction" at $-5\,^{\circ}\text{C}$ (L454–481); isotropy, and a collapse across porosity (L530–561); sintering "on the order of decades" (L596) and conductivity starting "near 0.0" (L611).

6 Conclusions

Draft: 5 Conclusion, treated as notes.

1. **What was done and found.** (new) [to write] One sentence of method, then the three results with their ranges: temperature rescales time (-40 to $-5\,^{\circ}\text{C}$, within 0.3%); conductivity follows $\text{SSA}^{-0.80}$ with the level set by porosity ($\varphi \leq 0.375$); vapor transport gives 41% of measured neck growth at $-20\,^{\circ}\text{C}$.
2. **What comes next.** (new) [to write] The laboratory test of Section 5.4; a second transport route in the model; three dimensions; colder temperatures and smaller grains.

Open Research; Acknowledgments

[to write] Code and data availability; run inventory (`RUN_TABLE.md`). Acknowledgments.

Supporting Information

- **Figures.** S1 domain-size convergence; S2 interface parameter; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Tables, text.** Complete parameters, if moved; packing construction; the weak-form derivation, if trimmed; run table.

Citations: what the .bib lacks

Suggested above and present in the `.bib`: all of them. Mentioned in this plan and *not* in the `.bib`:

- Auriault, Boutin & Geindreau (2009), the book whose naming of **t** the paper follows (doi:10.1002/9780470612033

- Ettrich et al. (2014), the tensor law applied to heat conduction; only if a second reference for the law is wanted.
- A source for packing and percolation thresholds of discs, if Section 5.3 states them.

Two keys are still in the old style in the `.bib` itself: `choukroun_strength_2020` and `nicoli_tensorial_2011`.

Smaller corrections throughout

- **Notation**, to the conventions listed at the top. Subscripts "ice" and "air" everywhere (ρ_{ice}, not ρ_i; ϕ_{ice}, ϕ_{air}); $\mathbf{K}_{\text{eff}}$ for the tensor and k_{eff} for its scalar mean, each used consistently; k_{ice}, k_{air} as in the tables; $\rho_{\text{air}}, P_{\text{atm}}$ in Eq. (2). ϕ_{air} is air, not water vapor.
- **Temperature.** Celsius with $T + T_0$ in Eqs. (3), (5); kelvin in Table 1. One convention.
- **Numbers.** 30 days (L172 says 28); mesh 2829×2829 (L362).
- **Words.** "Allen–Kahn"; "tonsorial"; "effect" for "affect"; "table table", "figure figure", "section section".